

Purva Pruthi

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SUMMARY

Machine learning researcher with a research focus on developing modular and scalable models that exhibit robust out-of-distribution generalization, causal reasoning capabilities, and interpretable, editable behavior.

Demonstrated impact by applying machine learning and causal modeling techniques to solve real-world decision-making problems across biology, finance, and materials science.

EDUCATION

University of Massachusetts Amherst 2018 – 2026 (Expected)

M.S./Ph.D. Candidate in Computer Science; GPA: (3.95/4.0)

Thesis: Compositional models for causal reasoning (Advisor: [David Jensen](#))

Indian Institute of Technology, Roorkee, India 2011 – 2015

Bachelor of Technology in Computer Science and Engineering; CGPA: (8.5/10.0)

EXPERIENCE

University of Massachusetts Amherst June 2020 – Present

Graduate Research Assistant

- Doctoral thesis focuses on combining the principles of causality, compositionality, and modular deep learning to develop efficient and scalable causal models of real-world systems.
- Systematic training and evaluation of the transformer architectures on [task-based compositional generalization](#) (NeurIPS 2026).
- Developed a novel [compositional framework for individual causal effect estimation](#) in diverse hierarchical systems ranging from SQL query execution engine to manufacturing assembly lines (CLEaR 2025).
- Improved evaluation methods for [observational causal inference](#) using experimental and empirical data sets (ICML 2021).
- Worked on DARPA's explainable AI (XAI) and competence aware machine learning (CAML) projects, developing techniques to explain the behavior of reinforcement learning agents and estimate their competence in novel settings.

Toyota Research Institute, Los Altos, USA June 2025 – August 2025

Research Intern – Energy and Materials Division

- Developed an LSTM-based time-series model to predict fuel cell performance from irregularly sampled time-series.
- Designed causal models using probabilistic programming framework [ChiRho](#) to estimate causal effects on voltage degradation.
- Applied a continuous optimization-based structure learning approach combined with prior knowledge to learn causal structure among material properties.

Google X, Mountain View, USA September 2021 – May 2022

Ph.D. AI Resident

- Developed predictive and causal modeling techniques for time-series forecasting of crop harvests in agriculture.
- Designed gene-gene interaction structure learning model for efficient design and editing of programmable plants for an early-stage biology project, now known as [Heritable Agriculture](#).

Amazon Research Center, Cambridge, UK May 2019 – August 2019

Research Intern – Supply Chain and Optimization

- Designed a modular [causal reinforcement learning](#) approach to enable efficient transfer learning in domains with perceptually changing state features but identical underlying causal dynamics model (ICML Workshop 2020).

Goldman Sachs, Bengaluru, India June 2015 – July 2018

Quantitative Analyst, Operations and Global Investment Research Division

- Designed and built infrastructure in C++ and R to backtest quantitative investment strategies based on company fundamentals and market data.
- Improved team efficiency by 25% by automating the reconciliation of external receipts and internal records using association rule mining.

SELECTED PUBLICATIONS

- [1] **Purva Pruthi**, Andrew Yuan, Alexander D’Amour, and David Jensen. “[Evaluating Compositional Generalization in Transformers: The Role of Composition Equivalence and Module Coverage.](#)” *Advances in Neural Information Processing Systems, Track on Evaluation and Datasets*. NeurIPS, 2026.
- [2] **Purva Pruthi** and David Jensen. “[Compositional Models for Estimating Causal Effects.](#)” *Causal Learning and Reasoning*. PMLR, 2025.
- [3] Amanda Gentzel, **Purva Pruthi**, and David Jensen. “[How and why to use experimental data to evaluate methods for observational causal inference.](#)” *International Conference on Machine Learning*. PMLR, 2021.
- [4] **Purva Pruthi**, Javier Gonzalez, Xiaoyu Lu, and Madalina Fiterau. “[Structure Mapping for Transferability of Causal Models.](#)” *Inductive Biases, Invariances, and Generalization in Reinforcement Learning Workshop, ICML 2020*.

SELECTED RESEARCH PROJECTS

Understanding compositional generalization in Transformers and LLMs March 2025 – Present

- Training of GPT-2 style transformer architectures and fine-tuning of large language models on the systematic train-test splits of code-reasoning tasks.
- Implemented transformer program-based architecture to generate mechanistically interpretable models on compositional reasoning tasks.

Single-cell modeling of breast cancer datasets Feb 2023 – Oct 2023

- Performed a systematic comparison of gene expression between healthy donors and cancer donors across two large-scale and sparse single-cell datasets: Human Breast Cell Atlas and Breast Cancer Atlas (100k cells).
- Performed various analyses – data integration, donor type prediction, DNA copy number variation analysis using a single-cell variational inference-based approach to distinguish gene markers among patients.
- Presented this work at CZI Single-Cell Biology 2023 Annual Meeting.

TECHNICAL SKILLS

Languages: Python, Pyro, C/C++, Java, R, SQL

Frameworks/Databases: PyTorch, TensorFlow, PostgreSQL, Git, Docker, HPC clusters (Slurm), Google Cloud, AWS

Developer Tools: Git, Docker, Visual Studio, PyCharm, Jupyter Notebook, RStudio, Jira

Domains: Causal inference, reinforcement learning, modular deep learning, Bayesian inference

HONORS AND AWARDS

UMass CICS Dissertation Writing Fellowship 2024

Data Science for the Common Good Fellowship May 2023 – August 2023

RELEVANT GRADUATE COURSEWORK

Machine Learning, Neural Networks: A Modern Introduction, Reinforcement Learning, Probabilistic Graphical Models, Distributed Operating Systems, Mathematical Statistics, Research Methods for Empirical Computer Science, Advanced Algorithms, Graph Theory

SERVICE AND OUTREACH

Reviewer/Program Committee: CLear’25, AAAI’24, AISTATS’24, AAAI’23, AISTATS’23, AISTATS’22 2022-2024

Data Science Industry Mentor for Chan Zuckerberg Initiative (CZI), Goldman Sachs 2023-2024

Mentor, Ph.D. Applicant Support Program, UMass CICS 2021

Mentor, EMBER Undergraduate Mentorship Program, UMass CICS 2021

Social Chair, UMass Graduate CS Women Group 2019-2020